COHEN BRYANT

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Portfolio: portfolio-six-ecru-11.vercel.app | LinkedIn: available on portfolio

EDUCATION

University of Wisconsin–Madison | *Madison, WI* | *Expected May 2030*
B.S. Mechanical Engineering | First-year undergraduate

TECHNICAL SKILLS

Engineering: SolidWorks (parts and assemblies), introductory FEA/CFD concepts, thermal and fluid-flow analysis, NASA CEA/CEARUN
Programming & Hardware: Python, MATLAB, C++, JavaScript, Arduino Nano ESP32, sensor integration, motor controls, Git
Fabrication: FDM/SLA 3D printing, manual lathe and mill, soldering, prototype assembly

ENGINEERING PROJECTS

Wisconsin Space Program (WISP) — Torch Ignitor Analysis | *Fall 2026–Present*

- Contributed to a collaborative GOX/propane torch ignitor engineering analysis, examining thermal response, material selection, injector-orifice sizing, and chamber-flow behavior.
- Used heat-transfer principles including the Bartz equation and Newton's law of cooling to compare material behavior against a six-second target firing duration.
- Organized and communicated findings using technical tables, graphs, and a two-page analytical presentation; full case study and PDF available on portfolio.

Transient Nuclear Reactor Xenon-135 Simulator | *Independent Project*

- Built an interactive JavaScript simulator of iodine-135/xenon-135 transient behavior following reactor power changes using coupled decay-kinetics equations.
- Developed visualizations of concentration response and reactor-control constraints; deployed interactive model at xenon-project-six.vercel.app.

ESP32 Wheeled Vehicle & Hardware Controls | *Hardware & Control Lead*

- Programmed Arduino Nano ESP32 firmware in C++ for motor-drive logic, speed control, and sensor signal processing.
- Modeled chassis and drivetrain components in SolidWorks for 3D printing; documented electrical connections and debugged firmware integration.

Moving Target Launcher — Mechanical Design & Trajectory Analysis | *Mechanical Design & Analysis*

- Created a physics-based trajectory model and designed an adjustable mechanical prototype; integrated machining and iterative physical testing.
- Compared launch testing with theoretical predictions, achieving reported agreement within 5% in the project documentation.

ADDITIONAL TECHNICAL PROJECTS

NucInfo Nuclear Systems Analytics — Developed a web dashboard to visualize nuclear energy metrics and comparative technical data. (nucinfo.vercel.app)
ME Connect — Created a deployed platform to connect engineering students with academic resources and project opportunities. (me-connect.vercel.app)

ACTIVITIES & EXPERIENCE

UW–Madison Engineering Organizations | *2026–Present*
WISP engineering analysis; involvement reported in AIAA, Liquid Rocketry, and American Nuclear Society activities.

Certified Nursing Assistant — North Central Health Care | *Aug 2024–Jul 2026*
Coordinated care with multidisciplinary teams in a time-sensitive setting; maintained clear communication and dependable task execution.

CERTIFICATIONS

Microsoft Office Specialist (2023) | MATLAB Onramp, MathWorks (2026)